# Phone-Watch Contrastive Representation Learning for Subject-Generalized Wearable Activity Recognition

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**Abstract** We study activity recognition from synchronized phone and watch inertial streams under a subject-disjoint evaluation protocol. The dataset provides per-subject files for phone and watch accelerometer and gyroscope recordings, with activity labels stored as contiguous letter-coded segments. We build a segment-aware preprocessing pipeline that pairs the four streams by activity episode, resamples them onto a shared 20 Hz timeline, and writes normalized 3 s windows with 1 s stride, yielding 50,091 training, 12,744 validation, and 47,790 test windows as $(12, 60)$ tensors.

We compare three tiers of learning: (i) classical models (Decision Tree, Gaussian NB, Linear SVM, Random Forest, AdaBoost, XGBoost) trained on 84-dimensional handcrafted feature vectors; (ii) supervised 1D CNN baselines trained on fused, watch-only, and phone-only raw windows; and (iii) a phone-watch contrastive model that learns paired embeddings via symmetric InfoNCE loss, evaluated through frozen linear probes.

The best current deep model is the supervised watch-only baseline, which reaches 63.55% accuracy and 0.6391 macro F1 on held-out subjects. The strongest contrastive result is the watch-only frozen probe at 58.31% accuracy and 0.5777 macro F1, while paired phone-watch probes improve over naive fusion but do not surpass the watch-only supervised model.

A consistent finding is strong device asymmetry: watch-only models substantially outperform phone-only models, suggesting that wrist motion carries most of the discriminative signal and that current phone-watch fusion is not yet robust enough to exploit both devices effectively.

## 1 Introduction

Human activity recognition from wearables is a natural test bed for representation learning: labels are expensive, while unlabeled sensor streams are plentiful. This project focuses on a phone-watch setting in which two devices observe the same activity from different body locations.

The working hypothesis is that synchronized phone and watch windows provide complementary views of the same underlying motion pattern. Aligning those views during pretraining should yield representations that transfer better to unseen subjects and under limited labels.

This report has two goals: to document a reliable preprocessing and windowing pipeline for the four sensor streams, and to compare watch-first supervised baselines against cross-view contrastive pretraining followed by frozen linear probes.

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## 2 Dataset Exploration

### 2.1 Raw Dataset Structure

The raw data are organized into subject-disjoint `train/` and `test/` folders, each containing four sensor collections: `phone/accel`, `phone/gyro`, `watch/accel`, and `watch/gyro`. Every file covers one subject and one stream; each row contains `subject_id`, an activity label, a timestamp, and three sensor axes. Labels are exposed as letter codes rather than human-readable activity names.

The cached window manifest contains 18 activity codes: A, B, C, D, E, F, G, H, I, J, K, L, M, O, P, Q, R, and S. All per-class analysis in this draft uses those codes directly, as the semantic mapping is not yet stored in the repository.

### 2.2 Split Summary

The raw paired-segment index contains 628 candidate activity episodes; 625 have all four required streams and 3 are dropped due to missing phone accelerometer and gyroscope data.

After preprocessing, we obtain the subject-disjoint split in Table 1. Validation subjects are deterministically carved from the raw training subjects, ensuring no leakage across splits.

| Split | # Subjects | # Windows | Notes |
|---|---|---|---|
| Train | 16 | 50,091 | Used for supervised fitting and contrastive pretraining. |
| Val | 4 | 12,744 | Held out from training subjects for model selection. |
| Test | 14 | 47,790 | Unseen subjects only; subject 1629 contributes a second activity occurrence per class. |

Table 1: **Window cache summary.** Cached windows under `artifacts/windows/` are the source of truth for all experiments.

The split is close to class-balanced: validation and test have 708 and 2,655 windows per class, respectively. Training is nearly balanced, with small deficits for B, F, J, and O due to the three removed episodes.

### 2.3 Dataset Summary

## 3 Annotation and Signal Exploration

Each file is an ordered concatenation of contiguous activity episodes rather than one continuous recording, and the same label can appear multiple times for one subject. We therefore index episodes by `(subject_id, activity_label, occurrence_index)` rather than by absolute timestamp. This segment-aware view is the key design choice that keeps phone-watch alignment faithful to the annotation structure.

Figure 1 illustrates the annotation structure directly. For subject 1600, both phone and watch accelerometer streams show long contiguous runs with synchronized A→B→C transitions. This justifies treating activity labels as episode-level segments rather than isolated frame-wise events.

Valid paired episodes are highly regular, with a mean overlapping duration of approximately 179.71 s and very small spread across subjects and classes. With a 3 s window and 1 s stride, each standard episode yields 177 windows.

Native sampling rates differ across streams: phone accelerometer averages 25.55 Hz, phone gyroscope 21.14 Hz, and both watch streams approximately 20.79 Hz. This heterogeneity makes direct sample-by-sample fusion inappropriate without resampling.

Figure 2 shows one representative paired segment for subject 1600, activity A. The top panel gives the raw phone/watch $x$-axis traces over the episode, and the bottom panel shows the selected 3 s window after interpolation to the shared 20 Hz grid.

| Statistic | Value | Notes |
|---|---|---|
| Participants | 34 | 20 raw training subjects and 14 raw test subjects. |
| Raw files | 136 | One file per subject-device-stream combination. |
| Input streams | 4 | Phone accelerometer, phone gyroscope, watch accelerometer, and watch gyroscope. |
| Activity codes | 18 | Letter-coded labels A–S, excluding N. |
| Raw sensor rows | 9,841,341 | Sum across all inertial files before segment pairing and windowing. |
| Days per participant | not directly observable | Each participant contributes exactly one raw file per stream in one split, which suggests one recording session per participant, but the timestamps are not linked to calendar dates. |
| Demographic metadata | unavailable | No age or gender fields appear in the shared dataset files. |

Table 2: **Raw dataset summary.** This table addresses the metadata items directly observable from the released files.

| Label | Rows | Label | Rows | Label | Rows |
|---|---|---|---|---|---|
| A | 555,095 | B | 532,870 | C | 538,369 |
| D | 546,977 | E | 558,964 | F | 530,915 |
| G | 549,077 | H | 544,383 | I | 546,991 |
| J | 531,125 | K | 568,730 | L | 542,065 |
| M | 553,721 | O | 540,826 | P | 551,872 |
| Q | 552,736 | R | 547,014 | S | 549,611 |

Table 3: **Raw sensor rows per activity code.** Counts are nearly balanced even before window extraction, ranging from 530,915 (F) to 568,730 (K).

## 4 Signal Preprocessing

The preprocessing pipeline parses each stream into contiguous runs of the same activity code; each run becomes one candidate segment with start row, end row, duration, and native sampling-rate metadata.

Four-stream episode pairs are built by matching segments with the same activity code and occurrence index. Only episodes containing all four streams (phone accelerometer, phone gyroscope, watch accelerometer, and watch gyroscope) are kept.

Within each valid episode, we compute the common overlapping duration across the four streams and operate on relative time within that overlap. Each sensor axis is linearly interpolated onto a shared 20 Hz grid. The final fused tensor stacks streams in a fixed order and produces a normalized $(12, 60)$ window. Channel-wise z-score statistics are estimated on the training windows and stored in `artifacts/windows/manifest.json`.

No additional low-pass or handcrafted denoising filter is applied in the current artifact set. This is a deliberate scope choice: the present report isolates the value of segment-aware alignment, interpolation, and normalization before introducing heavier preprocessing.

## 5 Windowing Strategies

Each window spans 3.0 s with a 1.0 s stride, yielding 60 samples after resampling. Window centers range from 1.5 s to 177.5 s within a standard episode. Because windows are generated after segment pairing, every window belongs to a single activity episode and never crosses label transitions—labels are inherited from the paired episode, and all cached windows are aligned

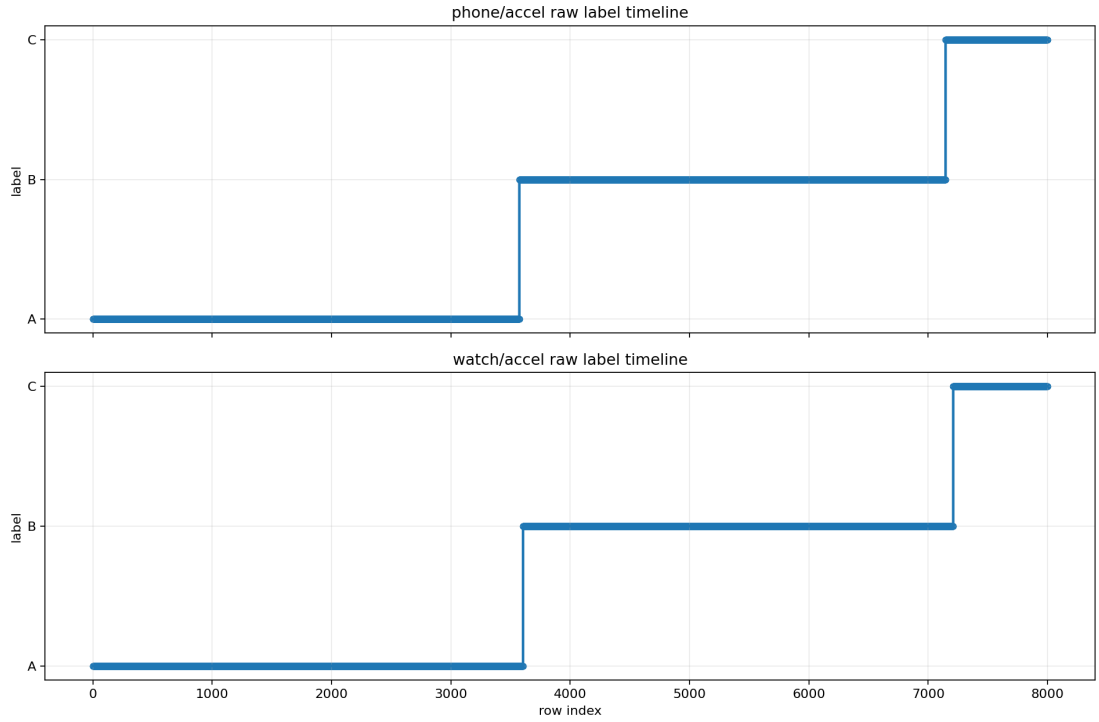



Figure 1: **Raw label timeline around labeled activity events (subject 1600).** The first 8,000 rows of the phone and watch accelerometer streams exhibit matching contiguous label blocks, including clear A→B→C transitions.

Figure 2: **Raw-to-resampled signal walkthrough (subject 1600, activity A).** The highlighted region in the raw segment becomes one 3 s analysis window after interpolation onto the shared 20 Hz timeline. This figure serves both as signal exploration and as a before/after preprocessing comparison.

across phone and watch.

For label-efficiency experiments, the 10% subset is selected deterministically within each

`(subject, class)` bucket, keeping at least one example per bucket when possible. This ensures that the 10% supervised baseline and the 10% probe use comparable labeled subsets.

## 6 Feature Extraction & Analysis

### 6.1 Handcrafted Features for Classical Baselines

We extract an 84-dimensional feature vector from each normalized $(12, 60)$ window. The 12 channels are ordered as phone accelerometer $(x, y, z)$, phone gyroscope $(x, y, z)$, watch accelerometer $(x, y, z)$, and watch gyroscope $(x, y, z)$. For each channel we compute four time-domain and three frequency-domain statistics, giving $12 \times 7 = 84$ features per window.

**Time-domain** (4 per channel): mean, standard deviation, root mean square (RMS), and zero-crossing rate (ZCR).

**Frequency-domain** (3 per channel): dominant frequency (Hz, from the peak of the one-sided FFT magnitude, excluding DC), spectral energy ($\sum |F_k|^2/N$), and spectral entropy ($-\sum p_k \log_2 p_k$, where $p_k$ is the normalized power spectrum).

Figure 3 shows the top-20 features ranked by Random Forest impurity importance. Watch accelerometer channels dominate, consistent with the watch-over-phone advantage observed across all model tiers.

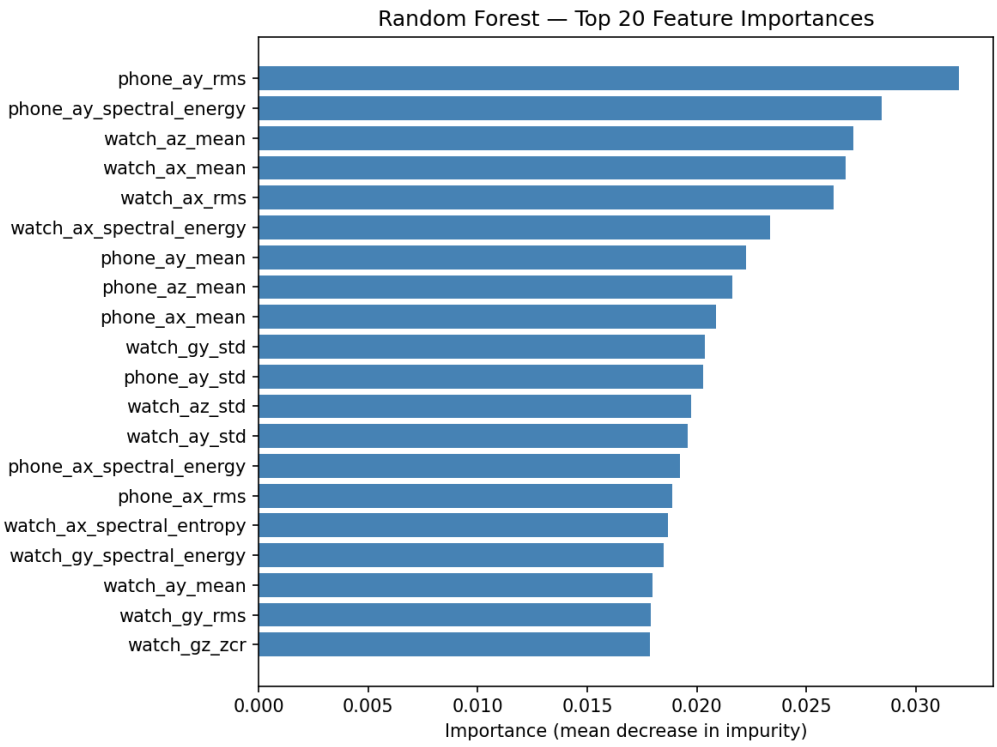



Figure 3: **Top-20 feature importances from Random Forest.** Watch accelerometer features rank highest, corroborating the device asymmetry finding.

### 6.2 Learned Representations

The supervised baseline uses a 1D CNN encoder (`TimeSeriesEncoder`) with three temporal convolution blocks, batch normalization, ReLU activations, and max pooling, followed by global average pooling. The encoder outputs a 256-dimensional feature vector; a linear classifier maps it to one of the 18 activity codes. Figure 4 shows the full architecture.

The contrastive model uses separate phone and watch encoders with the same backbone, each

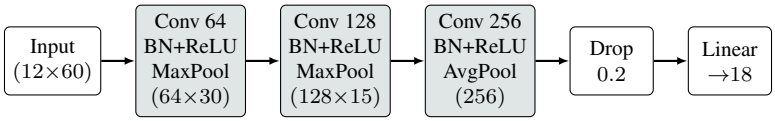



Figure 4: **Supervised HAR model.** The `TimeSeriesEncoder` backbone (three Conv1D blocks) is shared across all supervised and contrastive variants. Input channels are 12 for fusion mode, 6 for phone-only or watch-only.

consuming a $(6, 60)$ view and producing a 256-dimensional representation. Projection heads reduce embeddings from 256 to 128 to 64 dimensions before the symmetric InfoNCE loss is applied. Figure 5 illustrates the full pretraining setup and how the frozen encoder outputs are reused for linear probing.

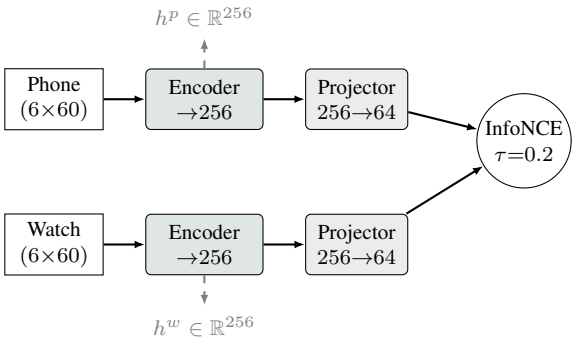



Figure 5: **Contrastive pretraining architecture.** Phone and watch views pass through separately parameterised encoders with the same backbone. After pretraining the encoders are frozen; $h^p$ and $h^w$ (dashed) are concatenated into a 512-D vector for the paired linear probe, or used individually for single-device probes.

For downstream evaluation, the paired probe concatenates frozen $h^p$ and $h^w$ into a 512-dimensional feature vector fed to a linear head. Single-device probes use only $h^p$ or $h^w$, isolating each device's contribution.

## 7 Modeling and Results

### 7.1 Classical Machine Learning

We train six classical models on the 84-dimensional handcrafted features: Decision Tree (max depth 20), Gaussian Naive Bayes, Random Forest (200 trees), Linear SVM with Platt calibration, AdaBoost (100 estimators), and XGBoost (200 trees, histogram method). All models are evaluated on the subject-disjoint test set—a more stringent protocol than random 80/20 or 70/30 splits, since test subjects are completely unseen during training. Table 4 reports macro precision, macro recall, accuracy, macro F1, and macro one-vs-rest AUC.

| Method | Prec. | Recall | Acc. | Macro F1 | Macro AUC |
|---|---|---|---|---|---|
| Decision Tree | 0.4331 | 0.4099 | 0.4099 | 0.4088 | 0.6934 |
| Gaussian NB | 0.5580 | 0.5447 | 0.5447 | 0.5354 | 0.8985 |
| Linear SVM | 0.5971 | 0.5833 | 0.5833 | 0.5662 | 0.9255 |
| Random Forest | 0.6221 | 0.6021 | 0.6021 | 0.5838 | 0.9503 |
| AdaBoost | 0.3999 | 0.3554 | 0.3554 | 0.3058 | 0.8894 |
| Hist. Grad. Boost | 0.5877 | 0.5727 | 0.5727 | 0.5549 | 0.9140 |

Table 4: **Classical ML results on the subject-disjoint test set.** Because the test split is perfectly class-balanced, macro recall numerically matches accuracy to four decimals. Random Forest achieves the best accuracy (60.2%) and macro AUC (0.950).

Figure 6 shows the confusion matrix of the strongest classical model, Random Forest. Its diagonal is sharp for classes B, O, P, Q, and S, while confusion remains concentrated among

the harder mid-performing classes such as D, F, H, I, J, K, and L. This error pattern is broadly consistent with the deep-model analysis later in the report.

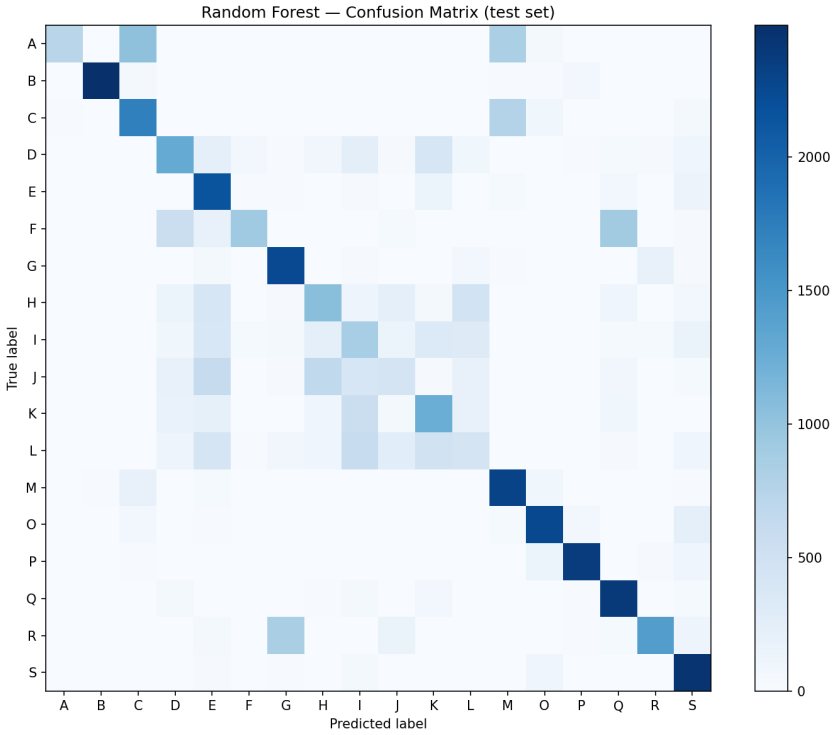



Figure 6: **Random Forest confusion matrix on the subject-disjoint test set.** The strongest classical baseline shows clean separation for several classes but still struggles on the same ambiguous activity groups seen in the deep models.

### 7.2 Supervised and Contrastive Setup

The supervised baseline is trained end-to-end with cross-entropy loss using AdamW (learning rate $10^{-3}$, weight decay $10^{-4}$, batch size 256, dropout 0.2) with early stopping on validation macro F1.

Contrastive pretraining uses training windows without labels. Phone and watch views are augmented with Gaussian jitter, mild amplitude scaling, and short temporal masking before being passed through the dual encoders. The pretraining objective is symmetric InfoNCE with in-batch negatives and temperature $\tau = 0.2$:

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^{N} \left[ -\log \frac{\exp(\mathrm{sim}(z_i^p, z_i^w)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(z_i^p, z_j^w)/\tau)} - \log \frac{\exp(\mathrm{sim}(z_i^w, z_i^p)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(z_i^w, z_j^p)/\tau)} \right]. \quad (1)$$

After pretraining, encoder weights are frozen and linear heads are trained for paired, phone-only, and watch-only probes at 10% and 100% label fractions.

### 7.3 Evaluation Protocol

All headline numbers are reported on the held-out test subjects. We track macro precision, macro recall, accuracy, macro F1, per-class F1, per-subject accuracy, confusion matrices, and ROC/AUC when probability outputs are available. Because the cached test split is exactly balanced at 2,655 windows per class, macro recall numerically equals overall accuracy in the tables below. The demo rubric mentions conventional random 80/20 or 70/30 splits, but we

instead report the stricter subject-disjoint protocol because unseen-subject generalization is the central project goal. The primary benchmark is now the watch-only supervised baseline, because it is the strongest simple model under subject-disjoint evaluation. Contrastive probes are therefore judged against the watch baseline first and against the weaker fusion and phone baselines second.

## 8 Results

### 8.1 Quantitative Summary

Table 5 summarizes results for the deep learning models. The strongest result is the full-label supervised watch baseline at 0.6355 accuracy, 0.6744 macro precision, 0.6355 macro recall, and 0.6391 macro F1. Even the regenerated 10%-label watch baseline remains strong at 0.6155 accuracy and 0.6078 macro F1, suggesting that the watch-only task is relatively label-efficient under the current windowing scheme.

The best contrastive result is the watch-only frozen probe at full labels, which reaches 0.5831 accuracy, 0.6033 macro precision, and 0.5777 macro F1. This does not exceed the supervised watch baseline, but it remains clearly stronger than the paired and phone-only frozen probes. Paired contrastive probes still outperform the older fusion baseline, but they remain well below watch-only supervision at both label budgets.

A notable result is that Random Forest (0.6021 accuracy, 0.6221 macro precision, 0.5838 macro F1, macro AUC 0.9503) remains competitive with the deep models. This is largely explained by device asymmetry: the current deep fusion setup is hampered by uninformative or unstable phone channels, whereas both the watch-only CNN and Random Forest can rely more heavily on wrist-dominated signals. Watch-centric features dominate the top-20 importances (Figure 3), confirming that wrist motion is the primary discriminative signal.

| Method | Input | Labels | Prec. | Recall | Acc. | Macro F1 |
|---|---|---|---|---|---|---|
| Supervised watch baseline | watch raw | 100% | 0.6744 | 0.6355 | 0.6355 | 0.6391 |
| Supervised watch baseline | watch raw | 10% | 0.6308 | 0.6155 | 0.6155 | 0.6078 |
| Contrastive watch probe | watch emb. | 100% | 0.6033 | 0.5831 | 0.5831 | 0.5777 |
| Contrastive watch probe | watch emb. | 10% | 0.5762 | 0.5647 | 0.5647 | 0.5556 |
| Contrastive paired probe | phone+watch emb. | 100% | 0.5138 | 0.4861 | 0.4861 | 0.4707 |
| Contrastive paired probe | phone+watch emb. | 10% | 0.5066 | 0.4795 | 0.4795 | 0.4658 |
| Contrastive phone probe | phone emb. | 100% | 0.1805 | 0.1692 | 0.1692 | 0.1540 |
| Contrastive phone probe | phone emb. | 10% | 0.1908 | 0.1793 | 0.1793 | 0.1603 |

Table 5: **Headline results.** The supervised watch baseline is the strongest deep model in the current artifact set. As in Table 4, macro recall equals accuracy because the test split is exactly class-balanced.

### 8.2 Interpretation

The dominant finding is device asymmetry. The watch-only supervised baseline reaches 0.6355 accuracy, far above the paired probe and any phone-only model. Random Forest's watch-centric importances reinforce this: top features are almost exclusively watch accelerometer channels, confirming that wrist motion is the cleanest source of discriminative information and that phone channels currently add noise rather than signal to fusion.

Paired contrastive pretraining is helpful relative to naive multimodal fusion, but it is not yet the best overall approach. The frozen watch probe comes closest to the supervised watch baseline, while the paired probe still trails substantially. This suggests that the current contrastive objective captures useful watch representations but does not yet make phone-watch combination reliably beneficial on unseen subjects.

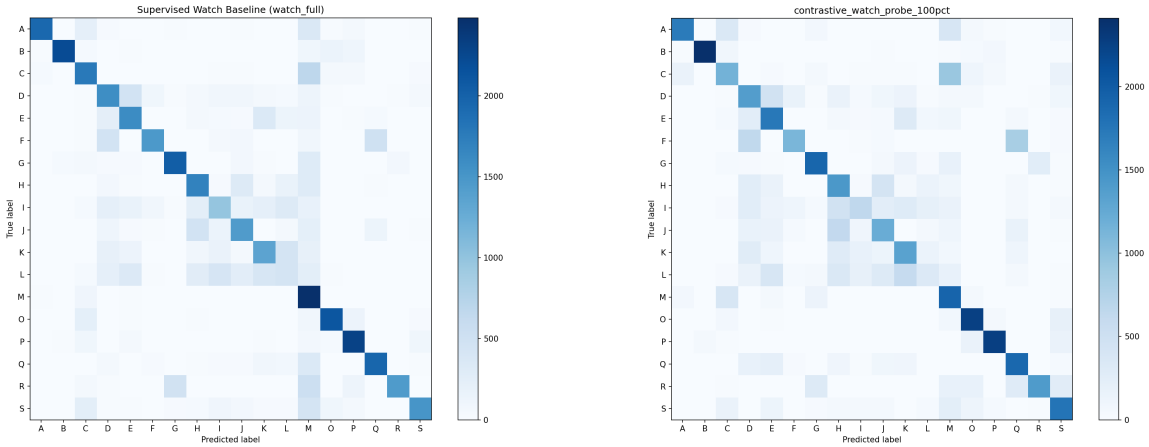
Per-class behavior is consistent across all three model tiers: classes B, O, and P are consistently easy, while F, I, J, K, and L remain difficult. Behavior-level interpretation is deferred until the

label-to-activity mapping is added to the repository.

An exploratory run with a lower learning rate reaches approximately 0.501 test accuracy and 0.486 macro F1 for the paired probe. We treat this as promising headroom rather than a main claim, as the standardized comparison table is based on the canonical artifact set and still falls short of the watch-only supervised benchmark.

### 8.3 Confusion Matrices

Figure 7 compares confusion matrices for the supervised watch baseline and the contrastive watch probe at full labels. Both exhibit the same broad pattern: easy classes (B, O, P) cluster on the diagonal while difficult classes (F, I, J, K, L) spread into off-diagonal bins. The supervised watch model shows consistently sharper diagonals, matching its stronger macro F1 in Table 5.




(a) Supervised watch (100% labels)

(b) Contrastive watch probe (100% labels)

Figure 7: **Confusion matrices on the subject-disjoint test set for the updated watch-first comparison.** Classes B, O, and P remain relatively easy, while F, I, J, K, and L remain challenging.

### 8.4 Per-Class F1 Comparison

Figure 8 shows per-class F1 for the supervised watch baseline and the contrastive watch probe. The clearest stable pattern is that classes B, O, and P are relatively easy, while F, I, J, K, and L remain difficult across both watch-based models. The supervised watch model is stronger on most classes, but the error profile is similar, suggesting that SSL is learning a related representation without yet overtaking direct supervision.

### 8.5 ROC / AUC

Figure 9 plots macro one-vs-rest ROC curves for the supervised watch baseline and the contrastive watch probe. Both curves lie well above the diagonal, but the supervised watch model achieves the stronger aggregate discrimination, consistent with its higher test accuracy and macro F1.

## 9 Conclusion

This project delivers a complete end-to-end pipeline from raw phone and watch inertial files to subject-disjoint quantitative evaluation. The main engineering contribution is a segment-aware alignment and resampling pipeline that converts heterogeneous raw streams into stable multimodal windows.

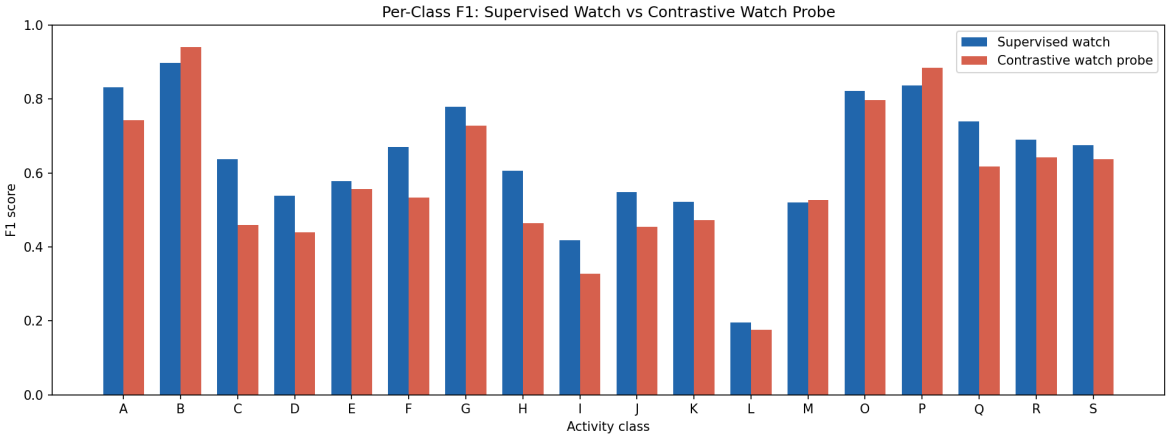



Figure 8: **Per-class F1 comparison across all 18 activity codes.** Supervised watch baseline (blue) and contrastive watch probe (red).

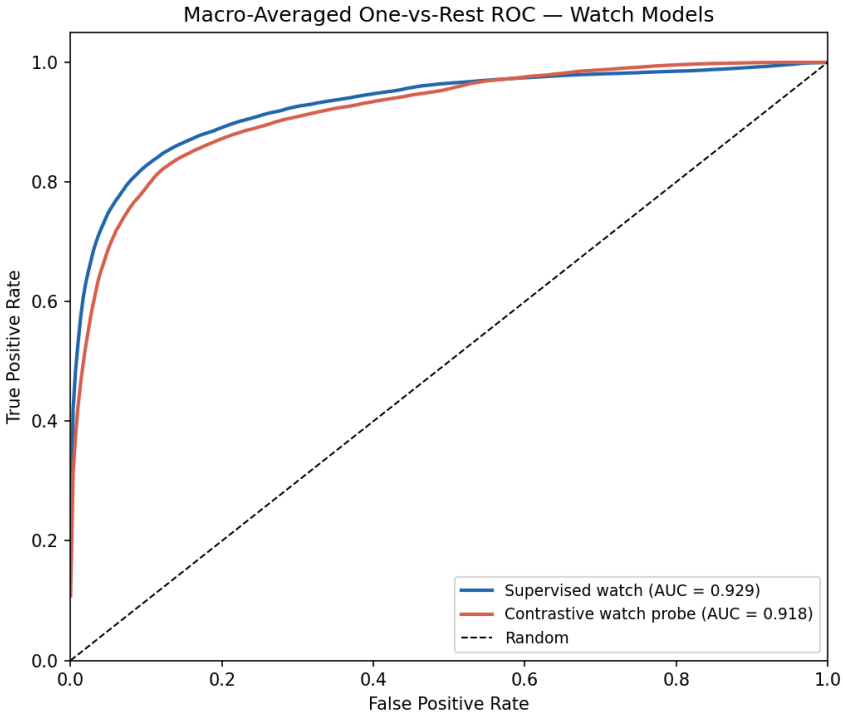



Figure 9: **Macro one-vs-rest ROC curves for the watch-first benchmark comparison.** Supervised watch baseline (blue) and contrastive watch probe (red).

The main empirical result is that the supervised watch-only baseline remains the best deep model under subject-disjoint evaluation, while the current contrastive pipeline does not yet surpass it. A complementary finding is strong watch-over-phone asymmetry that should shape the next round of analysis.

The three-tier evaluation still provides a useful ladder of baselines: classical models on handcrafted features establish an interpretable reference point, supervised watch training provides the strongest direct deep-learning benchmark, and contrastive pretraining shows partial gains over naive fusion without yet delivering the best overall result.

Next steps include understanding why watch-only models outperform fusion in the current scheme and testing whether stronger contrastive checkpoints or light fine-tuning can close the

remaining gap to the watch-only baseline.

**Rubric Alignment Notes**

The current artifact set covers dataset exploration, signal annotation, resampling and interpolation, windowing rationale, handcrafted features, classical and deep models, precision/recall/accuracy/macro-F1 reporting, confusion matrices, and ROC/AUC analysis. Two rubric items are not directly supported by the released files: demographic diversity and days recorded per participant, since the shared dataset contains only subject IDs, activity codes, timestamps, and inertial axes. Two additional items remain outside the present artifact set: explicit denoising filters and an exhaustive multi-window-size sweep. We therefore report them as limitations rather than claiming they were completed.

**Member Contributions**

Krishna Mula and Yash Khairnar contributed equally to this project. Work was shared jointly across dataset exploration, preprocessing and windowing, feature extraction, modeling and evaluation, figure generation, and report writing.

**Availability of Data and Software**

Raw inertial files are stored in `data/`. Derived window caches are in `artifacts/windows/`, supervised outputs in `artifacts/baseline/`, contrastive outputs in `artifacts/contrastive/`, and probe outputs in `artifacts/probes/`. Training and evaluation code is under `src/`; the execution workflow is documented in `docs/phase2_runbook.md`. Before public release, redistribution rights for the raw dataset should be confirmed.